

Problem statement :

Automatic Headlight Intensity Control using ESP32: Driving at night with high-intensity headlights can create glare for opposite drivers, reducing visibility and increasing the risk of accidents. Manual control of headlight brightness is not always effective and depends on driver attention. This project aims to design an ESP32-based automatic headlight intensity control system that adjusts headlight brightness automatically according to surrounding light conditions using an LDR sensor and PWM control.

Objectives:

The main objective of this project is to design and implement an ESP32-based automatic headlight intensity control system that ensures real-time brightness adjustment and improves driving safety.

- To detect ambient light intensity using an LDR sensor.
- To continuously monitor surrounding light conditions.
- To automatically control LED brightness using PWM.
- To reduce glare for opposite drivers during night driving.
- To improve visibility and driving safety.
- To develop a low-cost and energy-efficient lighting system.
- To understand real-time embedded system applications using ESP32

Design:

The proposed system is an ESP32-based Automatic Headlight Intensity Control System. The system is designed to monitor ambient light intensity and automatically adjust headlight brightness using PWM control. The design consists of three main parts: input layer (sensor), processing unit (ESP32), and output layer (LED headlights).

The design consists of three main parts:

1. Input Layer
2. Processing Unit
3. Output Layer

1. Input Layer (Sensor)

The system uses an LDR sensor to monitor surrounding light intensity.

- LDR Sensor Module: Detects ambient light intensity and provides analog output based on surrounding brightness.

The sensor continuously senses light levels and sends analog signals to the ESP32.

2. Processing Unit (ESP32)

The ESP32 acts as the central processing unit of the system.

- It reads analog data from the LDR sensor.
- It processes ADC values from GPIO34.
- It maps sensor values inversely to PWM duty cycle.
- It controls LED brightness using PWM output.

If bright light is detected → LED brightness decreases.

Else → LEDs remain at higher brightness.

3. Output Layer (LED Headlights)

The output devices perform actions based on ESP32 processing.

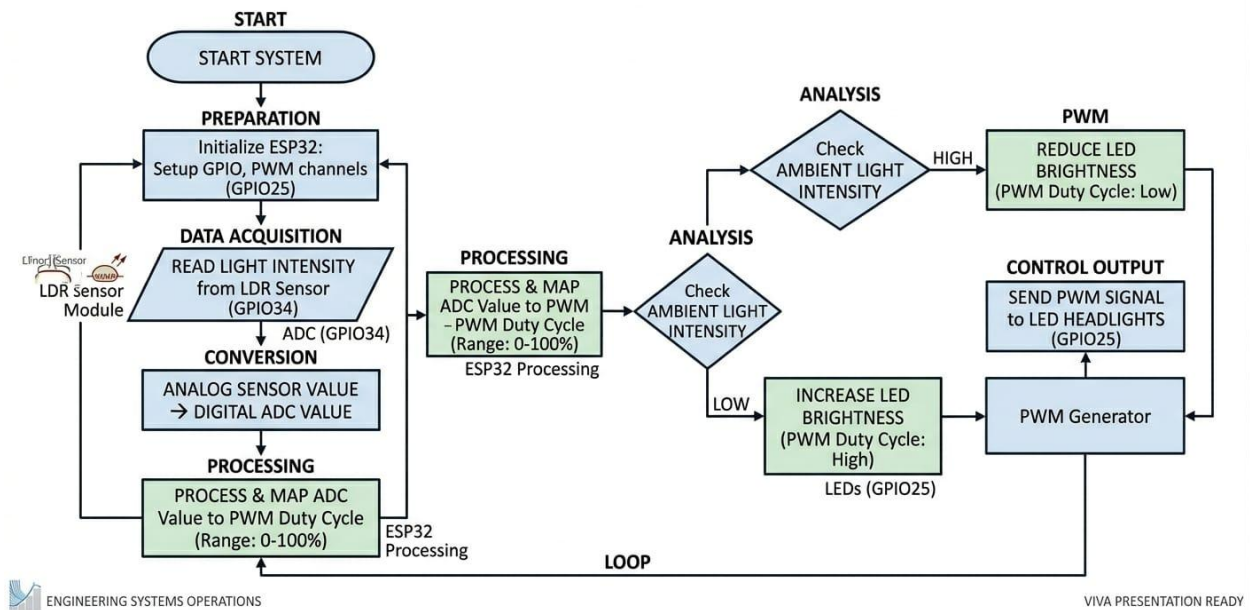
- LEDs: Act as vehicle headlights.
- PWM Output: Controls smooth brightness adjustment.

The LEDs automatically dim or brighten according to ambient light conditions.

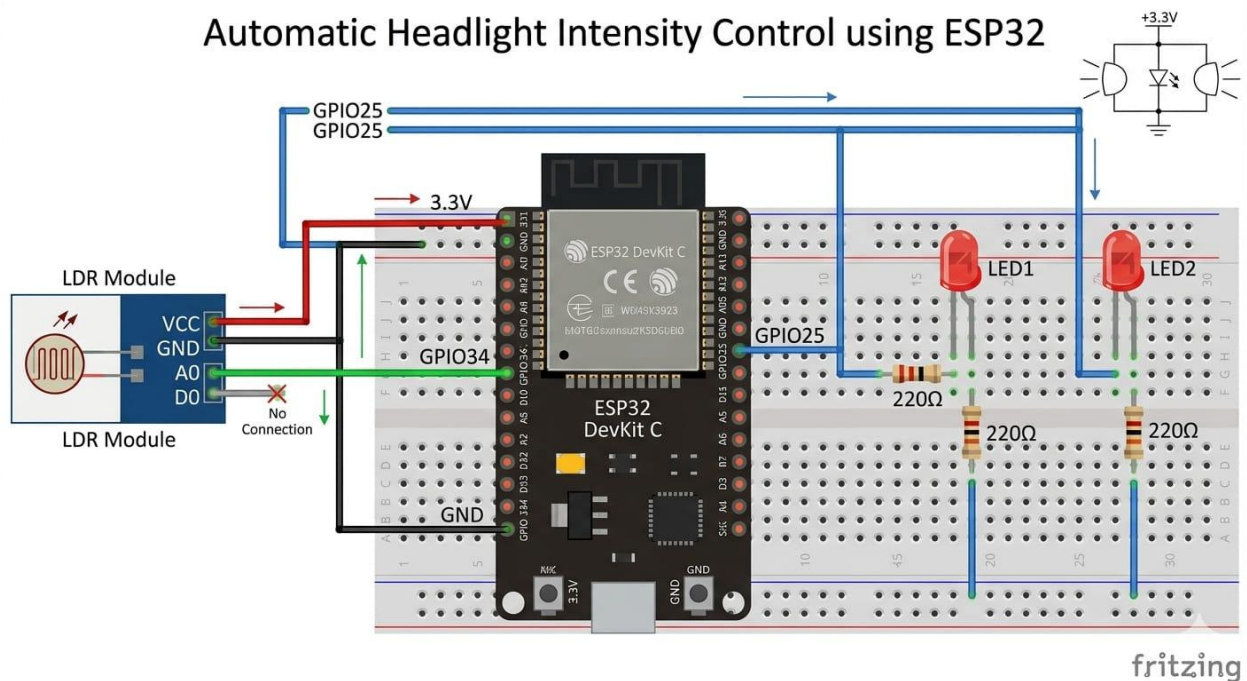
4. System Operation Flow

- LDR sensor continuously monitors ambient light.
- ESP32 reads analog sensor data.
- Sensor values are processed using ADC.
- PWM duty cycle is adjusted according to light intensity.
- LED brightness changes automatically.
- The system continuously operates in real time.

System Operation Flow – Automatic Headlight Intensity Control using ESP32



Automatic Headlight Intensity Control using ESP32



Tools/Components:

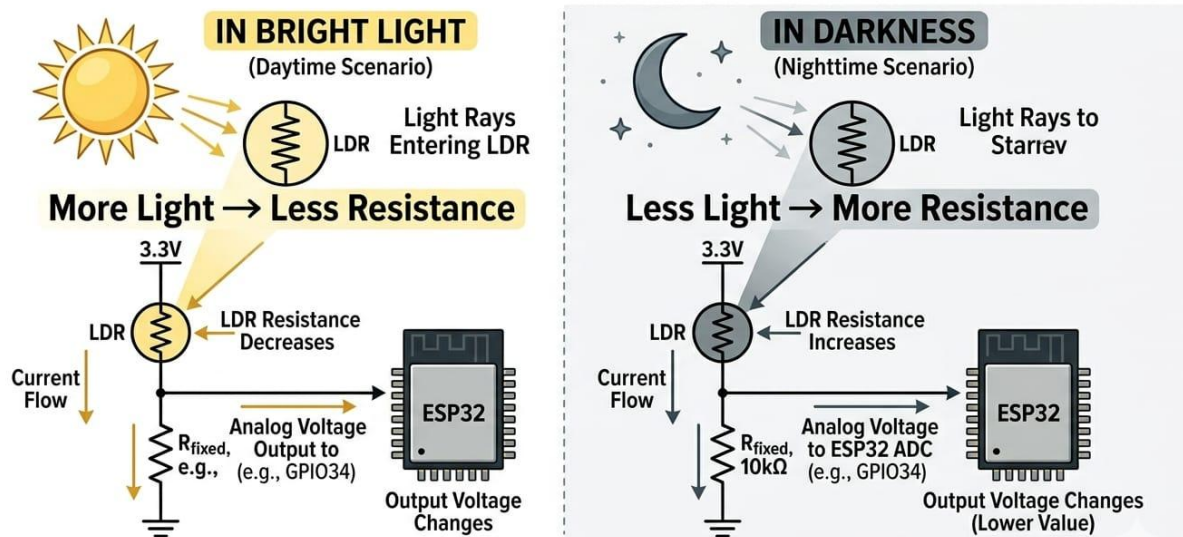
Hardware Components:

- **ESP32DevKit:**
Acts as the main controller of the system.
- **LDRSensorModule:**
Detects surrounding light intensity and provides analog output to ESP32.
- **Two 5mm LEDs:**
Used to represent vehicle headlights.
- **220 Ω Resistors:**
Used to limit current flowing through LEDs.
- **Breadboard:**
Used for temporary circuit connections.
- **Jumper Wires:**
Used to connect the ESP32, LDR sensor, and LEDs.
- **USB Power Supply:**
Used to power the ESP32 board.

Software Tools:

- **Arduino IDE:**
Used for writing and uploading the program to ESP32.
- **Embedded C Programming:**
Used to develop the control logic for reading sensor values and controlling LED brightness.
- **ESP32 Board Package:**
Required in Arduino IDE to program the ESP32 board.
- **Serial Monitor:**
Used to observe real-time LDR sensor values and PWM duty cycle during testing.

Working Principle of LDR



Circuit Operation:

The system is built around the ESP32 microcontroller, which interfaces with the LDR sensor module and LEDs. The LDR sensor senses the surrounding light intensity and sends an analog signal to the ESP32.

- The LDR sensor continuously detects ambient light intensity.
- The analog output of the LDR sensor is connected to GPIO34 of ESP32.
- ESP32 reads the sensor value using its ADC.
- The sensor value is mapped inversely to a PWM duty cycle.
- The PWM signal is given to LEDs connected to GPIO25.
- When the surrounding light is bright, the LED brightness is reduced.
- When the surrounding light is dark, the LED brightness is increased.
- The process runs continuously in a loop.

Working:

When the system is powered ON, the ESP32 initializes the LDR sensor input pin and the PWM output pin connected to the LEDs. After initialization, the system starts reading the surrounding light intensity continuously.

The LDR sensor senses the amount of light present in the environment. When more light falls on the LDR, its output changes accordingly. This change is converted into an analog voltage by the LDR sensor module. The analog voltage is then sent to the ESP32 through GPIO34, which is an ADC-supported input pin.

The ESP32 converts this analog voltage into a digital value using its internal ADC. This digital value represents the intensity of light detected by the LDR sensor. The ESP32 then maps this sensor value to a PWM duty cycle. Since the project uses inverse control, the LED brightness is reduced when bright light is detected and increased when the surroundings are dark.

PWM is used instead of simple ON/OFF control because it allows smooth variation of LED brightness. By changing the PWM duty cycle, the ESP32 controls how bright the LEDs glow. The LEDs act as vehicle headlights and automatically adjust their intensity according to the surrounding light conditions. This helps in reducing glare and improving night driving safety.

Challenges:

The following challenges were encountered during the design and implementation of Automatic Headlight Intensity Control using ESP32:

- **Sensor Calibration:**
The LDR sensor values may vary depending on room lighting, sunlight, and sensor position. Proper calibration is required for accurate brightness control.
- **Ambient Light Variation:**
Sudden changes in surrounding light may affect LED brightness quickly.
- **LDR Limitation:**
The LDR cannot distinguish between vehicle headlights, streetlights, and other light sources.
- **PWM Adjustment:**
Correct PWM mapping is required to achieve smooth dimming of LEDs.
- **ESP32 ADC Variation:**
ADC readings may fluctuate slightly due to noise and unstable light conditions.
- **Circuit Connection Issues:**
Loose jumper wires or incorrect resistor connections can affect LED operation.
- **Real Vehicle Implementation:**
For actual vehicle headlights, high-power LEDs or lamps require driver circuits instead of direct ESP32 connection.

Output:

The ESP32-based Automatic Headlight Intensity Control System was successfully implemented and tested.

- The LDR sensor successfully detected changes in surrounding light intensity.
- The ESP32 read analog values from the LDR sensor through GPIO34.
- PWM signal was generated successfully using ESP32.
- The brightness of LEDs changed automatically according to ambient light conditions.
- When bright light was detected, LED brightness decreased.
- When the surrounding area became dark, LED brightness increased.
- The system operated continuously in real time.
- Smooth brightness control was achieved using PWM.

Result :

- The project achieved automatic headlight intensity control.
- Real-time ambient light sensing was successfully implemented.
- PWM-based LED brightness control worked properly.
- The system helped demonstrate glare reduction during bright light conditions.
- A simple, low-cost, and energy-efficient smart lighting system was developed.

Applications:

The ESP32-based Automatic Headlight Intensity Control System can be used in various applications:

- **Smart Vehicle Lighting Systems:**
Used in vehicles to automatically adjust headlight brightness.
- **Automatic Street Lights:**
Used to control streetlight brightness based on surrounding light.
- **Intelligent Lighting Control Systems:**
Used in homes, offices, and industries for automatic brightness adjustment.
- **Energy-Saving Lighting Systems:**
Helps reduce power consumption by adjusting light intensity automatically.
- **Embedded System Learning Projects:**
Useful for understanding ADC, PWM, sensors, and ESP32 programming.
- **Automotive Safety Systems:**
Can be expanded for advanced vehicle headlight control.

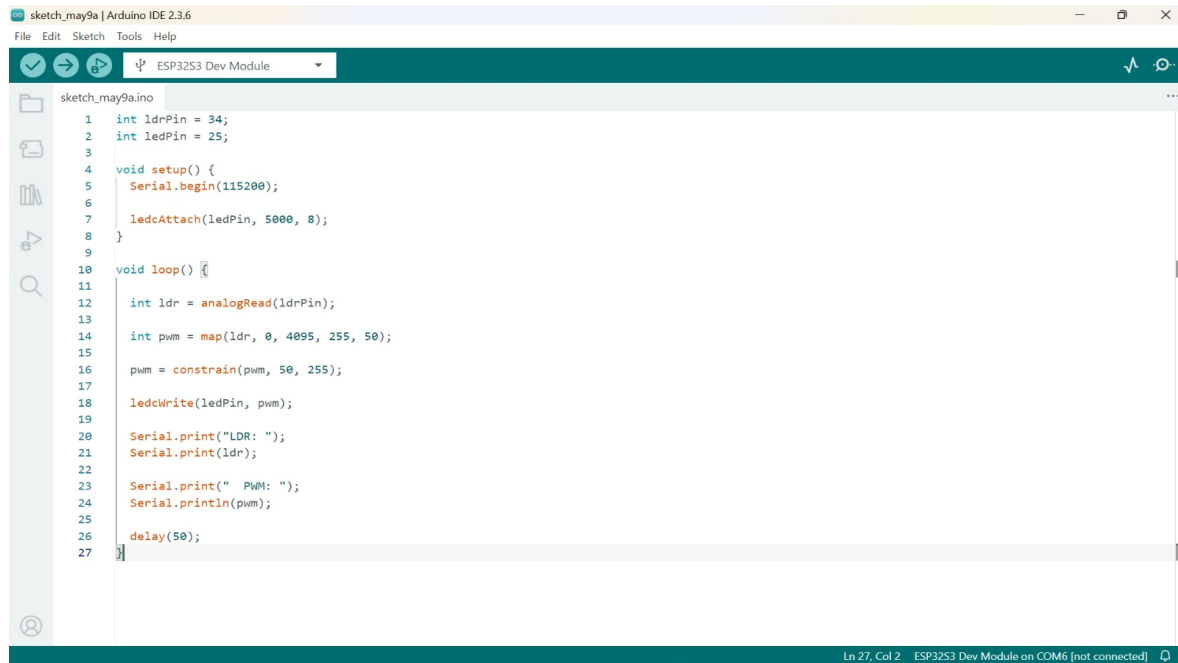
Conclusion:

The ESP32-based Automatic Headlight Intensity Control System was successfully designed and implemented. The system uses an LDR sensor to detect surrounding light intensity and controls LED brightness using PWM.

The ESP32 reads analog values from the LDR sensor and adjusts the brightness of LEDs automatically. When the surrounding light intensity increases, the LED brightness is reduced. When the surroundings become darker, the LED brightness increases. This helps reduce glare and improves night driving safety. The project demonstrates the practical use of ESP32, ADC, PWM, and sensor-based control in embedded systems. It is simple, cost-effective, and energy-efficient. The system can be further improved by using advanced sensors, real vehicle headlight drivers, and IoT-based monitoring.

Overall, the project proves to be an effective smart lighting solution and can be used as a basic model for automatic vehicle headlight control systems.

Demo:



```
sketch_may9a.ino
1  int ldrPin = 34;
2  int ledPin = 25;
3
4  void setup() {
5      Serial.begin(115200);
6
7      ledcAttach(ledPin, 5000, 8);
8  }
9
10 void loop() {
11
12     int ldr = analogRead(ldrPin);
13
14     int pwm = map(ldr, 0, 4095, 255, 50);
15
16     pwm = constrain(pwm, 50, 255);
17
18     ledcWrite(ledPin, pwm);
19
20     Serial.print("LDR: ");
21     Serial.print(ldr);
22
23     Serial.print("  PWM: ");
24     Serial.println(pwm);
25
26     delay(50);
27 }
```

Ln 27, Col 2 ESP32S3 Dev Module on COM6 [not connected]